

Solution Assignment 1 - MTH5011

Due Date: November 20, 2025

Total Marks: 20

Problem 1: Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear operator defined by

$$L \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ x + y \end{pmatrix}.$$

Find the standard matrix A representing L ; $A = (L(e_1), L(e_2))$.

Solution: The standard matrix A representing L is the 2×2 matrix whose columns are $L(e_1)$ and $L(e_2)$ respectively.

So,

$$L(e_1) = L \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 + 2(0) \\ 1 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Similarly,

$$L(e_2) = L \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 + 2(1) \\ 0 + 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Therefore,

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$$

Problem 2:

Determine whether or not the matrices

$$A = \begin{pmatrix} 3 & 2 \\ -1 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -1 & -2 \\ 2 & 1 \end{pmatrix}$$

commute with respect to multiplication, that is, whether $AB = BA$ or not.

Solution: We are given

$$A = \begin{pmatrix} 3 & 2 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & -2 \\ 2 & 1 \end{pmatrix}.$$

To check whether the matrices commute, we compute AB and BA .

$$AB = \begin{pmatrix} 3 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} -1 & -2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 3(-1) + 2(2) & 3(-2) + 2(1) \\ -1(-1) + 3(2) & -1(-2) + 3(1) \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ 7 & 5 \end{pmatrix}.$$

Now compute BA :

$$BA = \begin{pmatrix} -1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} -1(3) + -2(-1) & -1(2) + -2(3) \\ 2(3) + 1(-1) & 2(2) + 1(3) \end{pmatrix} = \begin{pmatrix} -1 & -8 \\ 5 & 7 \end{pmatrix}.$$

Clearly,

$$AB = \begin{pmatrix} 1 & -4 \\ 7 & 5 \end{pmatrix} \neq BA = \begin{pmatrix} -1 & -8 \\ 5 & 7 \end{pmatrix}.$$

Therefore, the matrices A and B do not commute.

Problem 3: Find an LU decomposition of the matrix

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Solution: We seek lower- and upper-triangular matrices L and U with unit diagonal for L such that

$$A = LU, \quad A = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Step 1. Write

$$L = \begin{pmatrix} 1 & 0 \\ \ell_{21} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix},$$

and impose $A = LU$.

Step 2. Multiply L and U :

$$LU = \begin{pmatrix} 1 & 0 \\ \ell_{21} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix} = \begin{pmatrix} u_{11} & u_{12} \\ \ell_{21}u_{11} & \ell_{21}u_{12} + u_{22} \end{pmatrix}.$$

Step 3. Equate LU to A :

$$\begin{pmatrix} u_{11} & u_{12} \\ \ell_{21}u_{11} & \ell_{21}u_{12} + u_{22} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

From the $(1,1)$ entry we get $u_{11} = 2$. From $(1,2)$ we get $u_{12} = 0$. From $(2,1)$ we get $\ell_{21}u_{11} = 1$, hence

$$\ell_{21} = \frac{1}{u_{11}} = \frac{1}{2}.$$

Finally, from $(2, 2)$:

$$\ell_{21}u_{12} + u_{22} = 2 \implies \frac{1}{2} \cdot 0 + u_{22} = 2,$$

so $u_{22} = 2$, and check shows

$$LU = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} = A.$$

Problem 4: Solve the system

$$10x_1 - x_2 = 25,$$

$$x_1 + 8x_2 = 43,$$

using Jacobi's method, with the initial guess $x^{(0)} = 0$, up to two iterations.

Solution: We have to find $x^{(1)}$ and $x^{(2)}$.

Rewriting for the Jacobi scheme:

$$x_1^{(k+1)} = \frac{25 + x_2^{(k)}}{10}, \quad x_2^{(k+1)} = \frac{43 - x_1^{(k)}}{8}.$$

First iteration:

$$x_1^{(1)} = \frac{25 + 0}{10} = 2.5, \quad x_2^{(1)} = \frac{43 - 0}{8} = 5.375,$$

so

$$x^{(1)} = \begin{pmatrix} 2.5 \\ 5.375 \end{pmatrix}.$$

Second iteration:

$$x_1^{(2)} = \frac{25 + 5.375}{10} = 3.0375, \quad x_2^{(2)} = \frac{43 - 2.5}{8} = 5.0625,$$

so

$$x^{(2)} = \begin{pmatrix} 3.0375 \\ 5.0625 \end{pmatrix}.$$

Thus, after two Jacobi iterations,

$$x^{(1)} = \begin{pmatrix} 2.5 \\ 5.375 \end{pmatrix}, \quad x^{(2)} = \begin{pmatrix} 3.0375 \\ 5.0625 \end{pmatrix}.$$